

Agent-X

Status Analysis

Where We Are, What We Need

*Graph inventory, formula coverage, NLP capabilities,
benchmark readiness, and roadmap.*

Graph nodes: 1,649 across 4 layers
Formulas: 23 mantras with automatic inversion
Tests: 329/417 passing (79%)
Benchmark coverage: 21% today → 93% with roadmap
NLP patterns: 16/23 working

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1. Architecture Summary

A one-page overview of how Agent-X works, for someone seeing the project for the first time.

What Agent-X Is

A proof-graph engine that answers physics and math questions by walking a 3D boolean tensor. No neural network, no LLM, no training data. All knowledge is a typed graph.

The 3D Tensor

$$G \in \{0, 1\}^{1649 \times 1649 \times 62}$$

1649 nodes (concepts, formulas, grammar), 62 relation types (identity, inverse, input-of, result-of, ...). Each edge is one cell set to 1.

The Pipeline

A question enters as a string and exits as a computed answer:

$$\text{String} \xrightarrow{\text{BQG}} \text{Triples} \xrightarrow{\text{refine}} \text{Triples} \xrightarrow{\text{expand}} \text{Triples} \xrightarrow{\text{detect}} \text{Signals} \xrightarrow{\text{dispatch}} \text{Answer}$$

1. **build-question-graph**: map each word to graph nodes (not string matching). Emit triples. 2. **assertion-bandha**: detect "X is Y" patterns, add swarupa edges. 3. **avrti-refine-v2**: bind numbers to concepts, propagate, fixpoint. 4. **grade-split**: separate sentences into graded ring grades. 5. **kosha-expand**: PPR walk to discover nearby concepts. 6. **detect-signals**: classify question type (derive, count, viveka, anumana). 7. **dispatch-answer**: find matching formula, compute (forward or inverse), format proof.

Key Properties

- **Monotonic**: triples only accumulate, never removed ($a \oplus b \geq a$) - **Mapping, not matching**: every word lookup is a graph traversal via naama edges - **Automatic inversion**: formulas inverted by reversing the krama chain using pratipaksha edges - **Composable**: tantras are declarative s-expressions, not imperative code - **Self-describing**: the pipeline's own stages are graph nodes

Implementation

Component	Technology	Lines
Engine	OCaml	6987
Knowledge graph	.om5 files (s-expression)	8743
Computation rules	.tantra4 files (s-expression)	2949
Word/metadata	.shabda files (key-value)	~3000
Analysis tools	Python (upakarana)	~5000
Tests	Python (pytest)	~3000

2. System Inventory

Current state of the Agent-X engine as of March 2026.

Engine

Component	Count	Description
OCaml modules	13	Core engine code
OCaml lines	6987	Total lines of OCaml
Graph nodes	1649	Knowledge nodes across 4 layers
Tantras	158	Declarative computation rules
Shabda nodes	1413	Word/metadata entries
Word mappings	783	Words mapped to graph nodes
Eval primitives	36	OCaml-backed operations
Relation types	62	Typed edge dimensions

Graph Layers

Layer	Nodes	Role
kosha	1007	Domain knowledge (physics, math, CS, biology)
sangati	346	Relations and connectors
bhasha	166	Language/grammar nodes
mantra	130	Formulas, operations, and computation rules

Edge Flow (Cross-Layer)

From / To	bhasha	kosha	mantra	sangati
bhasha	324	123	9	302
kosha	73	3621	169	2028
mantra	4	330	208	126
sangati	5	91	0	1882

Total graph edges: ~11,900. Density: 0.07% of possible cells in the 3D tensor.

Test Suite

Metric	Count
Total tests	417
Passing	329 (78.9%)
Expected failures (xfail)	81 (19.4%)
Actual failures	7 (1.7%)

Test Coverage by Layer

Layer	Tests
edges	79
grammar	45
gen_logic	43
physics	38
answers	24
pipeline	22
arithmetic	19
entity	19
evaluator	17
comparison	16
mixed	16
logic	28
graph	10
xfail	41

3. Formula Coverage (Mantras)

Agent-X has 23 physics/math mantras. Each mantra is a function stored in the graph with typed inputs (janya), output (phala), and a computation chain (krama).

All 23 Mantras

Mantra	Formula	Inputs → Output
kinetic-energy-mantra	$KE = \frac{1}{2}mv^2$	mass, velocity → kinetic-energy
potential-energy-mantra	$PE = mgh$	mass, gravity, height → potential-energy
momentum-mantra	$p = mv$	mass, velocity → momentum
newton-second-law-motion	$F = ma$	mass, acceleration → force
velocity-mantra	$v = u + at$	initial-velocity, acceleration, time → velocity
acceleration-mantra	$a = \frac{v-u}{t}$	initial-velocity, final-velocity, time → acceleration
pressure-mantra	$P = \frac{F}{A}$	force, area → pressure
mass-density-mantra	$\rho = \frac{m}{V}$	mass, volume → mass-density
work-mantra	$W = Fd \cos \theta$	force, displacement, angle → work
frequency-mantra	$f = \frac{1}{T}$	period → frequency
period-mantra	$T = \frac{2\pi}{\omega}$	angular-velocity → period
angular-velocity-mantra	$\omega = \frac{v}{r}$	velocity, radius → angular-velocity
angular-momentum-mantra	$L = I\omega$	moment-of-inertia, angular-velocity → angular-momentum
relative-velocity-mantra	$v_{rel} = v_{obs} - v_{observer}$	observed-velocity, observer-velocity → relative-velocity
spring-force-mantra	$F = kx$	spring-constant, displacement → spring-force
friction-force-mantra	$F = \mu N$	coefficient, normal-force → friction-force
centripetal-force-mantra	$F = \frac{mv^2}{r}$	mass, velocity, radius → centripetal-force
gravitational-force-mantra	$F = \frac{Gm_1m_2}{r^2}$	gravitational-constant, mass1, mass2, radius → gravitational-force
ohm-law	$V = IR$	current, resistance → voltage
electric-power-mantra	$P = VI$	voltage, current → electric-power
capacitance-mantra	$C = \frac{Q}{V}$	charge, voltage → capacitance
torque-mantra	$\tau = I\alpha$	moment-of-inertia, angular-acceleration → torque
photon-energy-mantra	$E = hf$	planck-constant, frequency → photon-energy

Inverse Computation

Every mantra with a krama chain supports automatic inversion. The pipeline reads each operation's inverse from the graph (pratipaksha edges on operation nodes):

Operation	Inverse	How
half ($x \times 0.5$)	double ($x \times 2$)	pratipaksha edge
square (x^2)	square-root ($\sqrt{x}$)	pratipaksha edge
multiplication ($x \times y$)	division (x/y)	pratipaksha edge
addition ($x + y$)	subtraction ($x - y$)	pratipaksha edge

No inverse formulas are hardcoded. The invert-krama tantra walks the krama tree and reverses each step using the graph.

Constants (Auto-Supplied)

Constant	Value	Used by
gravitational-acceleration	9.8	potential-energy-mantra, gravitational-force-mantra
planck-constant	6.626×10^{-34}	photon-energy-mantra
speed-of-light	299792458	(available, not yet used)
pi	3.14159265358979	period-mantra

Constants are stored as shabda entries on physics-constants and auto-supplied when a janya input has a constants-key in its shabda.

4. NLP Pattern Coverage

The pipeline parses natural language questions into triples via mapping (shabda-anveshana), not string matching. Here is what patterns work today and what does not.

Working Patterns (16/23)

Pattern	Example	Status
Direct: concept is N unit	"mass is 5 kg"	✓
Direct: concept is N (bare)	"force is 100"	✓
Separate sentences	"mass is 5 kg. velocity is 10 m/s."	✓
'of' pattern	"a ball of mass 5 kg"	✓
'with' pattern	"a ball with mass 5 kg"	✓
'has' pattern	"a ball has mass 5 kg"	✓
'has a X of N'	"has a mass of 5 kg"	✓
'equals' copula	"mass equals 5 kg"	✓
'the X is N'	"the mass is 5 kg"	✓
'given' prefix	"given mass is 5 kg"	✓
'what is' intent	"what is kinetic energy"	✓
'calculate' intent	"calculate kinetic energy"	✓
'determine' intent	"determine kinetic energy"	✓
'compute' intent	"compute kinetic energy"	✓
Inverse solve	"kinetic energy is 250. find velocity"	✓
Decimal values	"mass is 0.5 kg"	✓

Failing Patterns (7/23)

Pattern	Example	Why it fails	Impact
N before concept	"a 5 kg ball"	BQG expects concept then number	HIGH – very common in benchmarks
N unit concept	"5 kg mass"	Same – number before concept	HIGH
Variable = N	"m = 5 kg"	"m" maps to metre, "=" not copula	MEDIUM
Multi-step chain	"height is 10. find velocity"	No PE → KE conservation chain	HIGH
Pronoun reference	"it has velocity 10"	No coreference resolution	MEDIUM
Scientific notation (partial)	"9.109e-31"	Works for some, edge cases	LOW
Units inline (partial)	"5 kilogram"	Works sometimes, inconsistent	LOW

Morphological Support

shabda-anveshana applies plural → singular normalization via graph-stored rules:

Rule	Example	Method
english-plural-regular	"forces" → "force"	strip "s"
english-plural-ies	"energies" → "energy"	strip "ies", add "y"
english-plural-es	"masses" → "mass"	strip "es"

Rules are graph nodes (under vachana-bahu), not hardcoded. New rules = new nodes.

Intent Detection

5 dispatch modes, detected by signal analysis:

Mode	Trigger	Example
derive	vidhi-kaala + numbers	"mass is 5. find KE"
anumana	"what is X" without numbers	"what is momentum"
viveka	"is X a Y"	"is velocity a vector"
count	"how many"	"how many forces act"
shunya	no intent, no data	(empty question)

5. Benchmark Coverage

We evaluated Agent-X against two standard physics QA benchmarks: TheoremQA (TIGER-Lab, 2023) and PhysReason (2024).

TheoremQA

Source: TIGER-Lab/TheoremQA on HuggingFace. 800 questions across math, physics, CS, finance.

Filter	Count
Total questions	800
Physics (text-only, no images)	140
Covered by existing mantras	~95
Solvable with current NLP	~60

Questions by Formula Category

Category	Questions	Have Mantra?
velocity/speed	30	✓ velocity-mantra
electric/circuit	32	✓ ohm-law, electric-power, capacitance
frequency/wave	14	✓ frequency-mantra (partial)
thermal/heat	10	needs thermal-mantra
kinetic energy	8	✓ kinetic-energy-mantra
density	8	✓ mass-density-mantra
friction	8	✓ friction-force-mantra
gravity	7	✓ gravitational-force-mantra
acceleration	7	✓ acceleration-mantra
power	6	needs power-mantra (general)
angular	5	✓ angular-velocity-mantra
work	5	✓ work-mantra
spring	4	✓ spring-force-mantra
pressure	4	✓ pressure-mantra
momentum	2	✓ momentum-mantra

Coverage Gap

The 140 physics questions fall into three tiers:

Tier 1 – Directly solvable (~30 questions): Single-formula, values given explicitly. Example: "Calculate the momentum of an 80 kg person moving at 2 m/s." Our pipeline handles these today with minor NLP adjustments.

Tier 2 – Solvable with NLP fixes (~65 questions): Correct mantra exists, but values are embedded in natural language that our BQG cannot yet parse. Example: "A 5 kg ball is moving at 10 m/s. What is the kinetic energy?" Needs the N-before-concept pattern.

Tier 3 – Needs new mantras or multi-step (~45 questions): Requires formulas we do not have (thermal, wave, gas law, oscillation) or multi-step chains (conservation of energy).

PhysReason

Source: PhysReason on HuggingFace. 1200 physics problems, 147 theorems.

Filter	Count
Total problems	1200 (mini: 200)
Text-only (no images)	31 (mini)
Within our formula vocabulary	~15
Solvable with current NLP	~5

PhysReason problems are harder: multi-step, symbolic answers, require context understanding. These represent our long-term target.

GSM8K

8792 grade-school math problems. Zero physics formula questions. Not applicable.

Benchmark Readiness Score

Benchmark	Total applicable	Currently solvable	With NLP fixes	With new mantras
TheoremQA	140	~30 (21%)	~95 (68%)	~130 (93%)
PhysReason (mini)	31	~5 (16%)	~10 (32%)	~20 (65%)
Our own tests	417	329 (79%)	—	—

6. Missing Vocabulary and Mantras

To reach full benchmark coverage, we need to add these formulas and supporting nodes.

Priority 1: Missing Mantras (unlocks ~55 benchmark questions)

Mantra needed	Formula	Inputs → Output	Benchmark Qs
thermal-energy-mantra	$Q = mc\Delta T$	mass, specific-heat, temperature-change → heat	11
wave-speed-mantra	$v = f\lambda$	frequency, wavelength → wave-speed	9
power-mantra	$P = \frac{W}{t}$	work, time → power	9
shm-period-mantra	$T = 2\pi\sqrt{\frac{m}{k}}$	mass, spring-constant → period	7
pendulum-period-mantra	$T = 2\pi\sqrt{\frac{l}{g}}$	length, gravity → period	7
ideal-gas-mantra	$PV = nRT$	pressure, volume, moles, gas-constant → temperature	6
coulomb-mantra	$F = k\frac{q_1q_2}{r^2}$	charge1, charge2, distance → coulomb-force	5
electric-field-mantra	$E = \frac{F}{q}$	force, charge → electric-field	5
magnetic-force-mantra	$F = qvB$	charge, velocity, magnetic-field → magnetic-force	3
projectile-range-mantra	$R = \frac{v^2 \sin 2\theta}{g}$	velocity, angle, gravity → range	2
conservation-energy	$PE = KE$	via pratipaksha chain	1

Priority 2: Supporting Nodes Needed

Node	Type	Shabda needed
specific-heat	kosha	word: "specific heat"
temperature-change	kosha	word: "temperature change", "delta t"
wavelength	kosha	word: "wavelength"
wave-speed	kosha	word: "wave speed"
moles	kosha	word: "moles", "mol"
gas-constant	kosha	constants-key: gas-constant (8.314)
charge	kosha	word: "charge" (already exists?)
electric-field	kosha	word: "electric field"
magnetic-field-strength	kosha	word: "magnetic field"
range	kosha	word: "range"
length	kosha	word: "length"

Priority 3: Constants to Add

Constant	Value	Key in physics-constants
gas-constant	8.314	gas-constant
coulomb-constant	8.99×10^9	coulomb-constant
boltzmann-constant	1.381×10^{-23}	boltzmann-constant
electron-charge	1.602×10^{-19}	electron-charge

Priority 4: Conservation Chain (pratipaksha)

The pratipaksha edge already exists: kinetic-energy ↔ potential-energy. But dispatch-derive does not chain through pratipaksha to convert one to the other. Adding this unlocks "dropped from height, find velocity" questions.

Required: when solve-for has no direct mantra match, check if a pratipaksha concept has a matching mantra, compute that, then use pratipaksha to convert.

7. NLP Roadmap

The NLP gaps are more impactful than the mantra gaps. Fixing 3 patterns unlocks ~65 more benchmark questions with existing mantras.

Fix 1: Number-Before-Concept (HIGH - unlocks ~40 questions)

Current: "a 5 kg ball" → FAILS (5 binds to kilogram, not mass) Target: "a 5 kg ball" → [mass, sankhya, 5.] + [mass, matra, kilogram]

The fix is in build-question-graph's emit-triples: when a number+unit appears before a concept, do a forward-lookahead to find the concept and bind to it.

Pattern variations that should work after fix: - "a 5 kg ball" - "a 0.60 kg basketball" - "5 kg mass" - "the 100 N force" - "a 2 m height"

Fix 2: Period-After-Bare-Number (MEDIUM - unlocks ~15 questions)

Current: "area is 5." → "5" stays asprista-sankhya because "." consumed as viraam Target: "area is 5." → [area, sankhya, 5.]

The fix is in BQG's punctuation handling: when a period follows a digit, it should be treated as a number terminator first, sentence boundary second.

Fix 3: Multi-Step Chains via Pratipaksha (HIGH - unlocks ~10 questions)

Current: "height is 10. find velocity" → NO MATCH Target: compute PE = mgh, then KE = PE (conservation), then $v = \sqrt{2*KE/m}$

Requires chaining through pratipaksha edges in dispatch-derive. When solve-for (velocity) is a janya of mantra-A (kinetic-energy-mantra), and mantra-A's phala (kinetic-energy) has a pratipaksha (potential-energy) that IS the phala of mantra-B (potential-energy-mantra) whose janya inputs ARE bound: compute mantra-B first, assign result to kinetic-energy, then inverse-solve mantra-A.

Fix 4: Pronoun Resolution (MEDIUM - unlocks ~5 questions)

Current: "a ball weighs 5 kg. it has velocity 10 m/s." → 5 binds to pronoun-it Target: resolve "it" to the most recent entity (mass/ball)

Requires tracking the current entity in BQG and substituting pronoun-it with the antecedent.

Fix 5: Variable Names (LOW - unlocks ~5 questions)

Current: "m = 5 kg" → "m" maps to metre Target: "m" in context of "mass m = 5" should bind to mass

Requires context-aware disambiguation: when a single letter follows a concept name, treat it as a variable alias, not a word lookup.

Summary: Impact of Each Fix

Fix	Effort	Questions Unlocked	Cumulative Coverage
Baseline (today)	–	~30 / 140 (21%)	21%
+ Fix 1 (N before concept)	Medium	+40	50%
+ Fix 2 (period after number)	Small	+15	61%
+ Fix 3 (pratipaksha chain)	Medium	+10	68%
+ New mantras (11 formulas)	Medium	+35	93%
+ Fix 4 (pronouns)	Medium	+5	96%
+ Fix 5 (variables)	Small	+5	100%